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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

AD2310-1 – SOFTWARE TESTING AND AUTOMATION

(For AD)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which of the following is not a part of STLC?
A) Testing Planning
B) Requirement Gathering
C) Test Design
D) Testing closure
2. In which environment we can performed the Alpha testing?
A) User's end
B) Developer's end
C) User's and developer's end
D) Customers end
3. Which of the following is a static testing technique?
A) Functional Testing
B) Regression Testing
C) Code Reviews
D) Load Testing
4. What is a walkthrough in software testing?
A) A formal review conducted by developers
B) An informal review where a presenter leads team members through a document
C) A process of executing test cases
D) A type of performance testing
5. Which of the following is a test design technique?
A) Equivalence Partitioning
B) Unit Testing
C) Load Testing
D) Automation Testing
6. Which document is created during the Test Planning phase of STLC?
A) Test Summary Report
B) Test Strategy and Test Plan
C) Test Execution Report
D) Incident Report
7. What is the main purpose of effort estimation in test management?
A) To allocate sufficient time and resources for testing activities
B) To create defects in the software
C) To write the software code
D) To perform database backup
8. What is Testopia used for?
A) Code debugging
B) Test case documentation and test management
C) Performance testing
D) Software deployment
9. What is the first step in defect management?
A) Defect Logging
B) Defect Closing
C) Defect Retesting
D) Defect Verification
10. Who is responsible for fixing a defect in the defect lifecycle?
A) Testers
B) Developers
C) Project Managers
D) Business Analysts
11. Which of the following is a key challenge in test data management?
A) Ensuring test data security and compliance
B) Writing test cases faster
C) Automating test execution
D) Developing new software features

12. Which of the following is NOT a reason to automate testing?
 A) To reduce human errors
 B) To execute repetitive test cases faster
 C) To eliminate the need for test planning
 D) To improve test coverage
13. ITG stands for-----?
 A) Integration Testing Group
 B) Instantaneous Test Group
 C) Independent Test Group
 D) Individual Testing Group
14. What is the main limitation of automation testing?
 A) It is completely error-free
 B) It requires initial investment and maintenance
 C) It cannot execute repetitive tasks
 D) It eliminates the need for test scripts
15. Which component of Selenium is used for record-and-playback testing?
 A) Selenium WebDriver
 B) Selenium IDE
 C) Selenium Grid
 D) Selenium RC
16. Which of the following features does Selenium WebDriver support?
 A) Mobile application testing
 B) Desktop application testing
 C) Web application testing across multiple browsers
 D) Mainframe application testing
17. What is the main advantage of using Selenium Grid?
 A) It allows testing across multiple browsers and machines simultaneously
 B) It eliminates the need for writing test scripts
 C) It is only used for mobile testing
 D) It replaces the need for WebDriver
18. Which tool is commonly used for logging and tracking defects?
 A) Selenium
 B) Bugzilla
 C) TestNG
 D) JIRA
19. In Selenium, which command is used to launch a web browser?
 A) driver.get("URL")
 B) driver.open("URL")
 C) driver.load("URL")
 D) driver.navigate("URL")
20. What is JIRA primarily used for?
 A) Conducting API testing
 B) Writing test scripts
 C) Performing performance testing
 D) Managing software development projects and tracking bugs

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Distinguish between the root cause of a defect and its effects.	5	L*2	1	1
	b) Explain the differences between different types of review: informal review, technical review, walkthrough and inspection	6	L2	2	1
	c) Discuss the role of a tester in improving software quality. How does their role evolve during the testing life cycle?	5	L2	2	1
2.	a) Explain the key reasons for Testing. How it is necessary in software development?	8	L2	1	1
	b) Differentiate between Functional Testing and API Testing with suitable examples.	8	L2	2	1
3.	a) Explain the various stages of the Software Testing Life Cycle. What document templates are generated at each stage?	8	L2	1	1
	b) What are different types of Specification-Based techniques in Testing? A hospital diagnoses patients based on symptoms and test results:				

- i. If the patient has a high fever, positive test results, and difficulty breathing, they are diagnosed with Severe Infection.
- ii. If the patient has a high fever and positive test results but no difficulty breathing, they are diagnosed with a Mild Infection.
- iii. If the patient has a high fever but negative test results, they are diagnosed with a Common Cold.
- iv. If the patient has no fever, they are considered Healthy.

8	L2	2	1
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Unit – II

4. a) Describe the process of test case documentation. How does it contribute to test management?
- b) Interpret the role of Test Execution in defect management.
- c) Describe the Defect Life Cycle with a diagram.

5	L2	3	1
5	L2	5	1
6	L2	5	1

5. a) Develop a sample test data set in MS Excel for testing the registration functionality of a web application. Include positive and negative test cases.
- b) What steps would you follow to close a defect after it is fixed? Discuss the role of regression testing in this process.

10	L2	3	1
6	L2	5	1

6. a) How does Testopia help in test case documentation and test management?
- b) What are the different test management activities? Explain any three in detail.

6	L2	3	1
10	L2	3	1

Unit – III

7. a) List and describe any four popular tools used in automation testing. How do these tools differ in terms of features and use cases?
- b) Create a simple automation test case using Selenium WebDriver and TestNG for testing a login form. Include all necessary steps from setting up the environment to running the test.
8. a) Write a Selenium WebDriver script to automate the login functionality of a web application using TestNG. Include necessary validations in the test script.
- b) Explain the advantages and disadvantages of automation testing compared to manual testing.

8	L2	4	1
8	L3	4	1
8	L3	4	1
8	L2	4	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
